CAPSTONE PROJECT REPORT: BRAIN-TASKS CLOUD-NATIVE INFRASTRUCTURE

Capstone Project Report: Brain-Tasks Cloud-Native Infrastructure

Prepared by: Bibek Kumar Sahu

Project Name: Brain Tasks: Managed DevOps Lifecycle

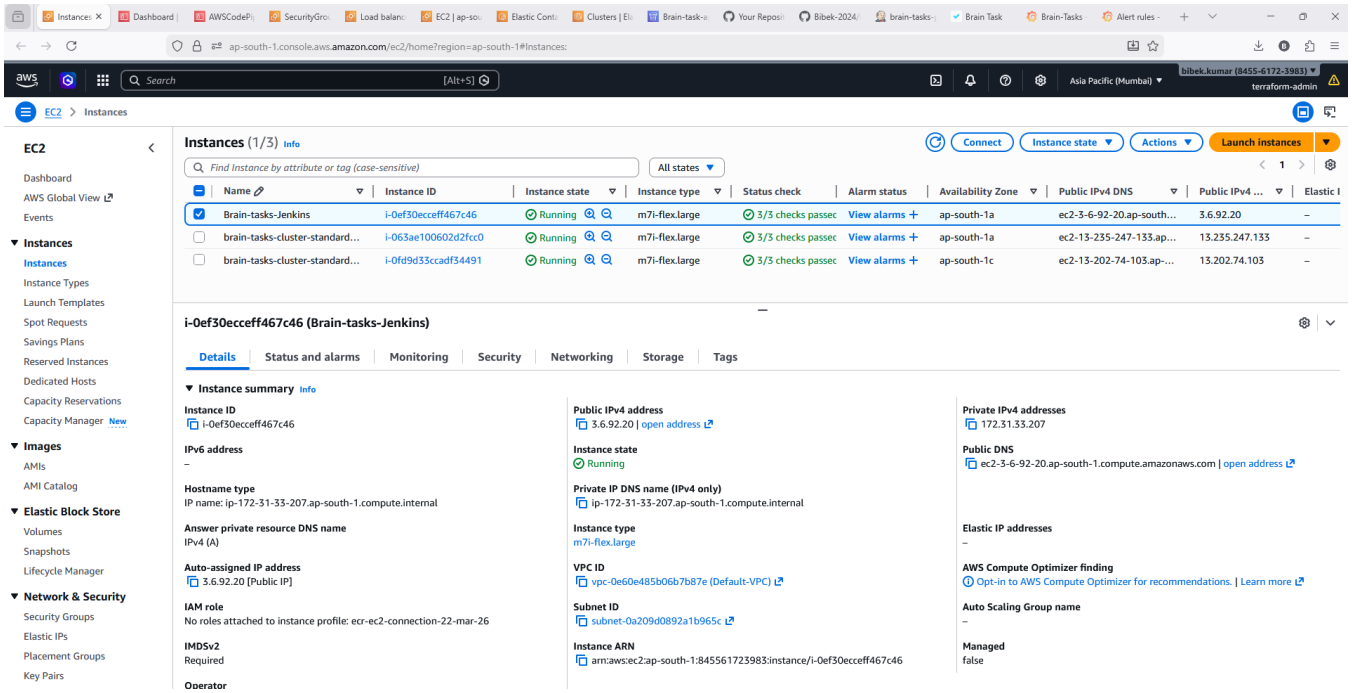
Status: Production-Ready & Monitored

Technology Stack: AWS (EKS, ECR, EC2, CodePipeline), Jenkins, Docker, Kubernetes, Prometheus, Grafana.

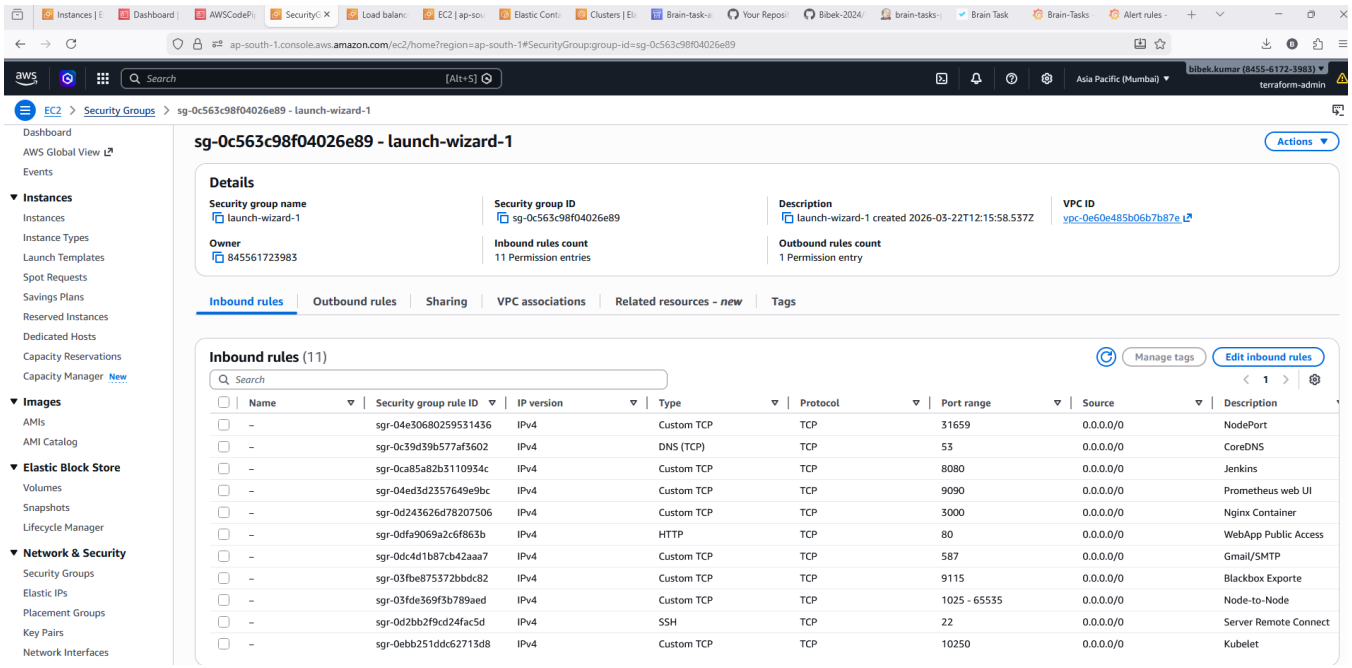
I. Infrastructure & Security Foundation

The project begins with establishing a secure, scalable AWS environment. All compute and networking resources were configured to allow seamless communication between the CI/CD tools and the Kubernetes cluster.

- **[1.1] Compute Resources:** Deployment of the **Jenkins Master Node** and **EKS Worker Nodes** on m7i-flex.large instances for high-performance builds.



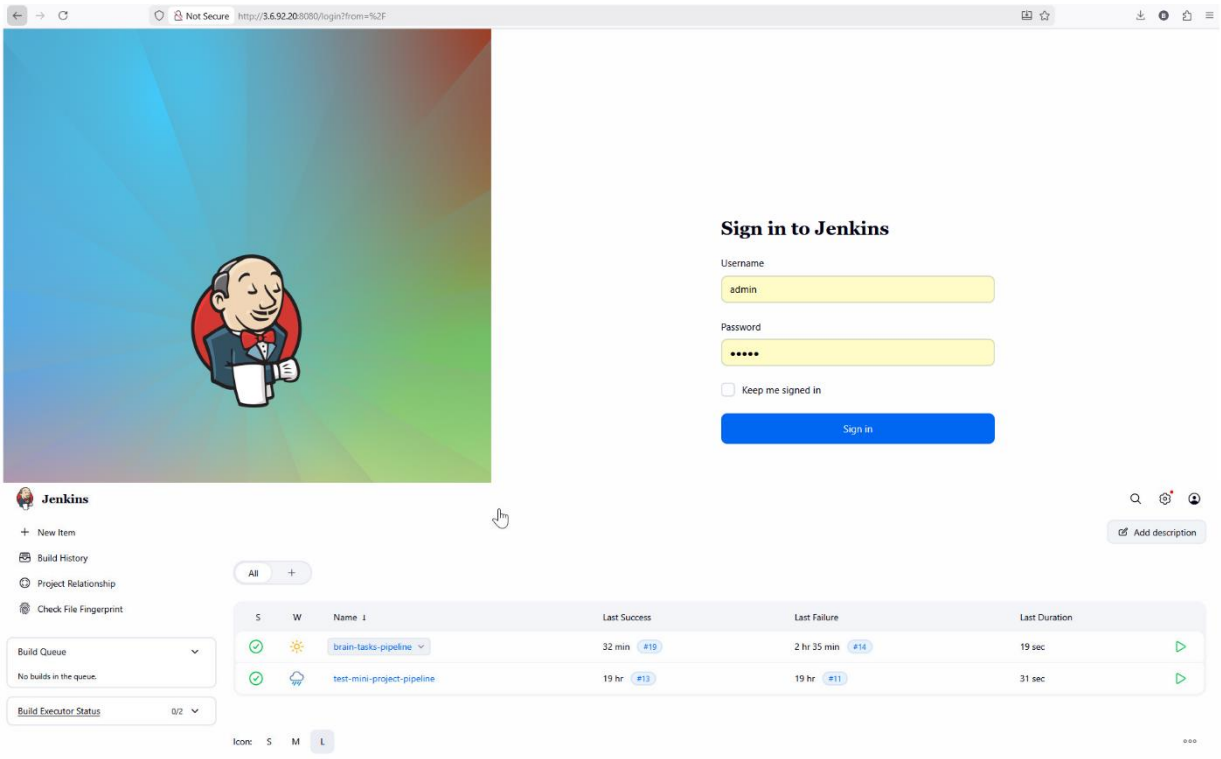
- **[1.2] Security Group Configuration:** Implementation of a strict firewall (Security Group launch-wizard-1) opening only essential ports: 8080 (Jenkins), 3000 (Grafana), 9090 (Prometheus), and 80 (WebApp).



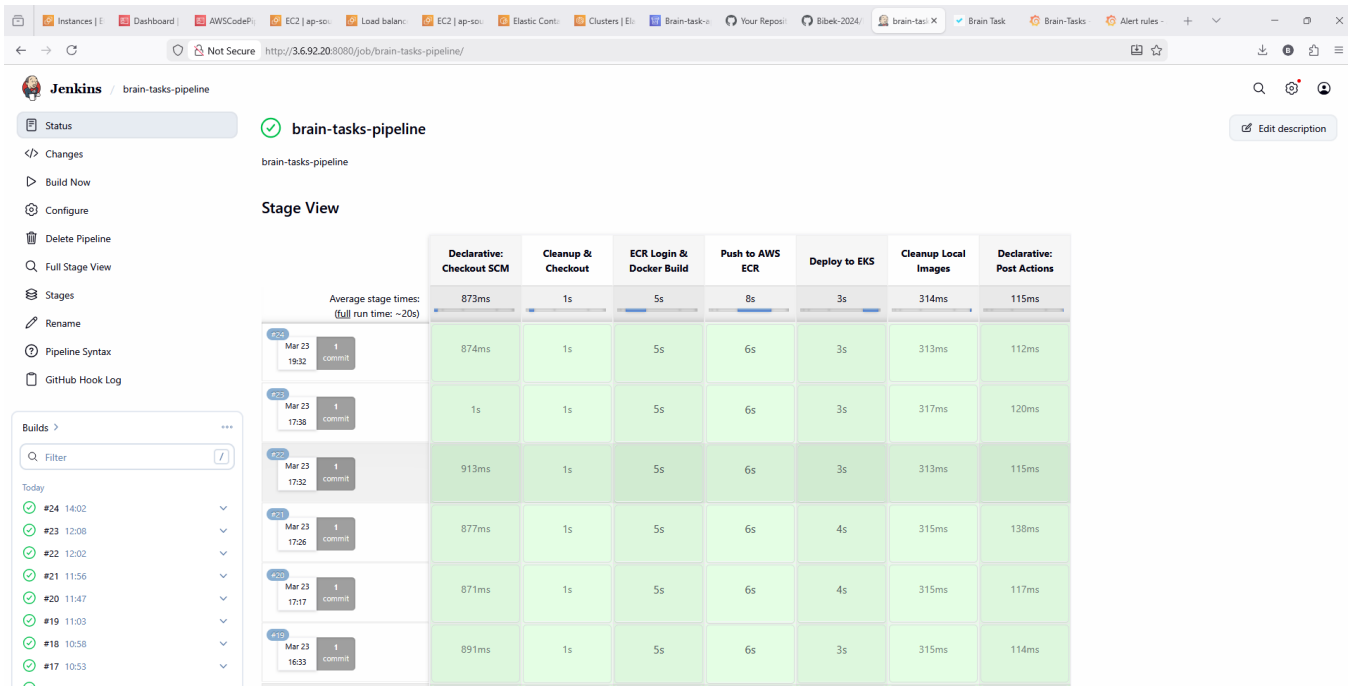
II. CI/CD Pipeline Orchestration

The heart of the project is the automated pipeline that moves code from GitHub to the cloud without manual intervention.

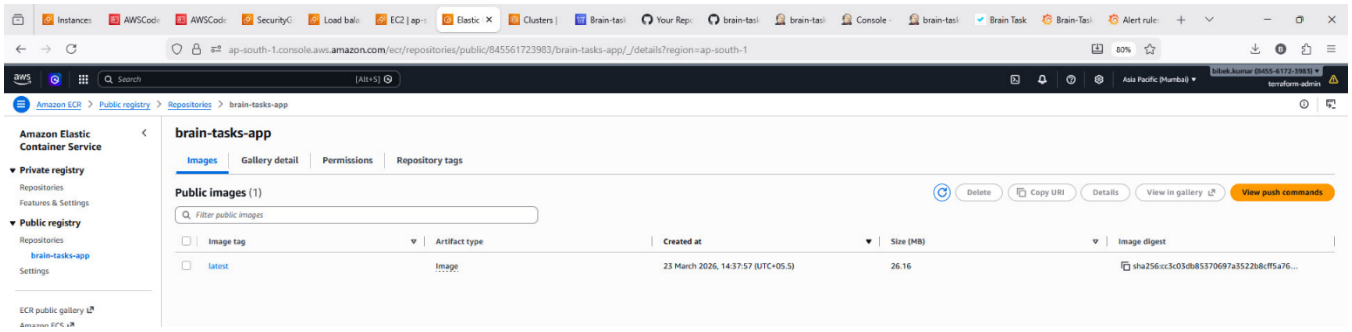
- **[1.3] Jenkins Automation:** The Jenkins dashboard manages the brain-tasks-pipeline. It handles the Docker build process and communicates with AWS ECR.



- **[1.4] Pipeline Stage View:** A detailed 7-stage pipeline was implemented, covering:
    1. **SCM Checkout** (GitHub)
    2. **Docker Build**
    3. **ECR Push**
    4. **EKS Deployment**
    5. **Cleanup**



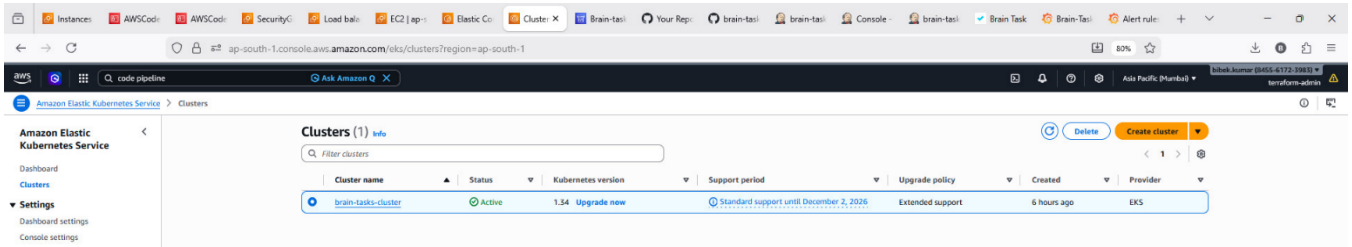
- **[1.5] Container Registry (AWS ECR):** Successful push of the latest build image to a private AWS Elastic Container Registry, ensuring image versioning and security.



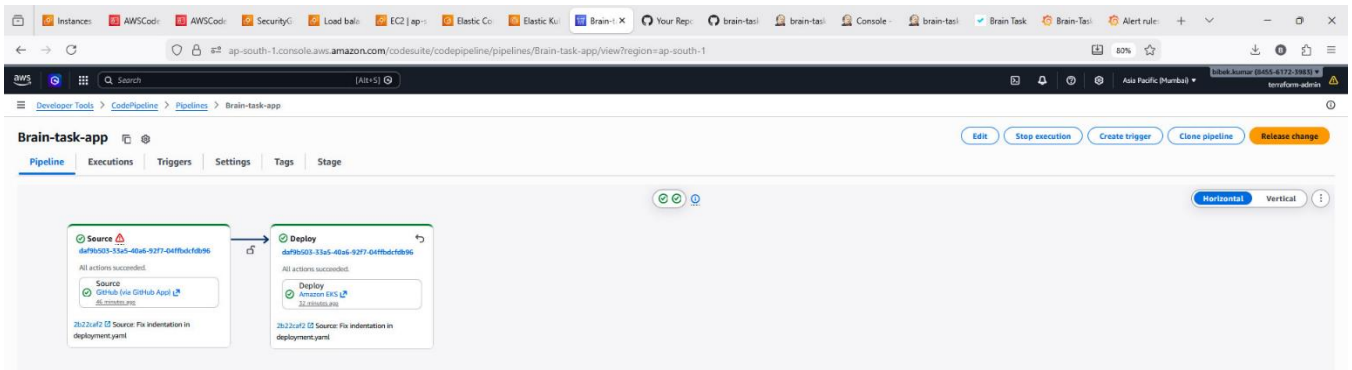
III. Cloud-Native Deployment (AWS EKS)

The application is hosted on a managed Kubernetes cluster to ensure high availability and auto-scaling.

- **[1.6] EKS Cluster Management:** The brain-tasks-cluster is running on Kubernetes version 1.34 in the ap-south-1 region.



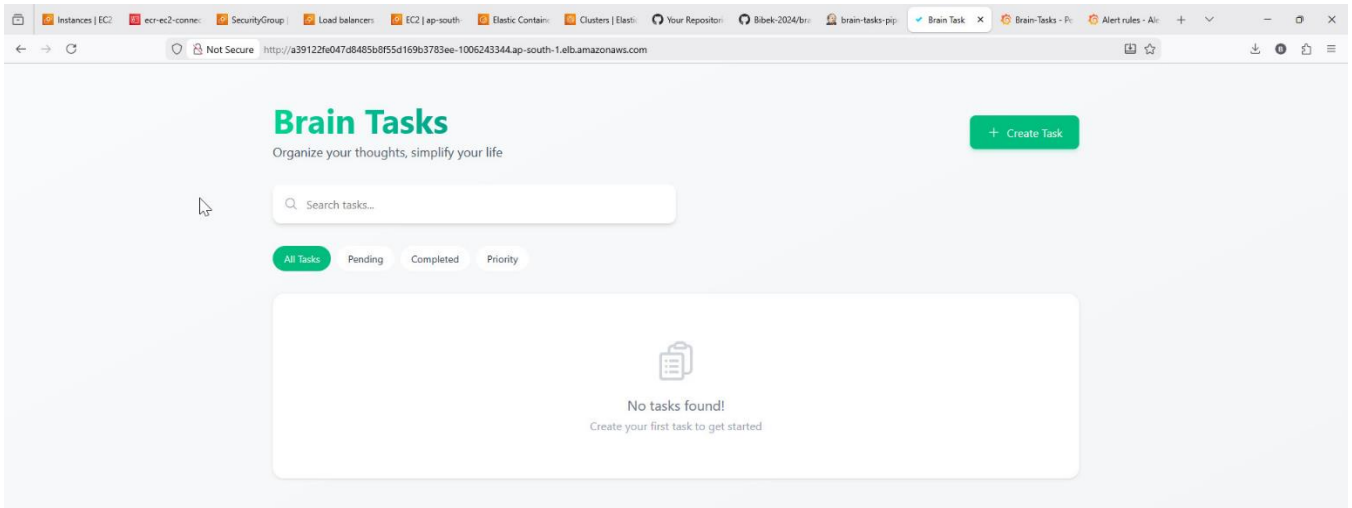
- **[1.7] CodePipeline Integration:** AWS CodePipeline acts as the final deployment trigger, bridging the gap between Jenkins-built images and the EKS environment.



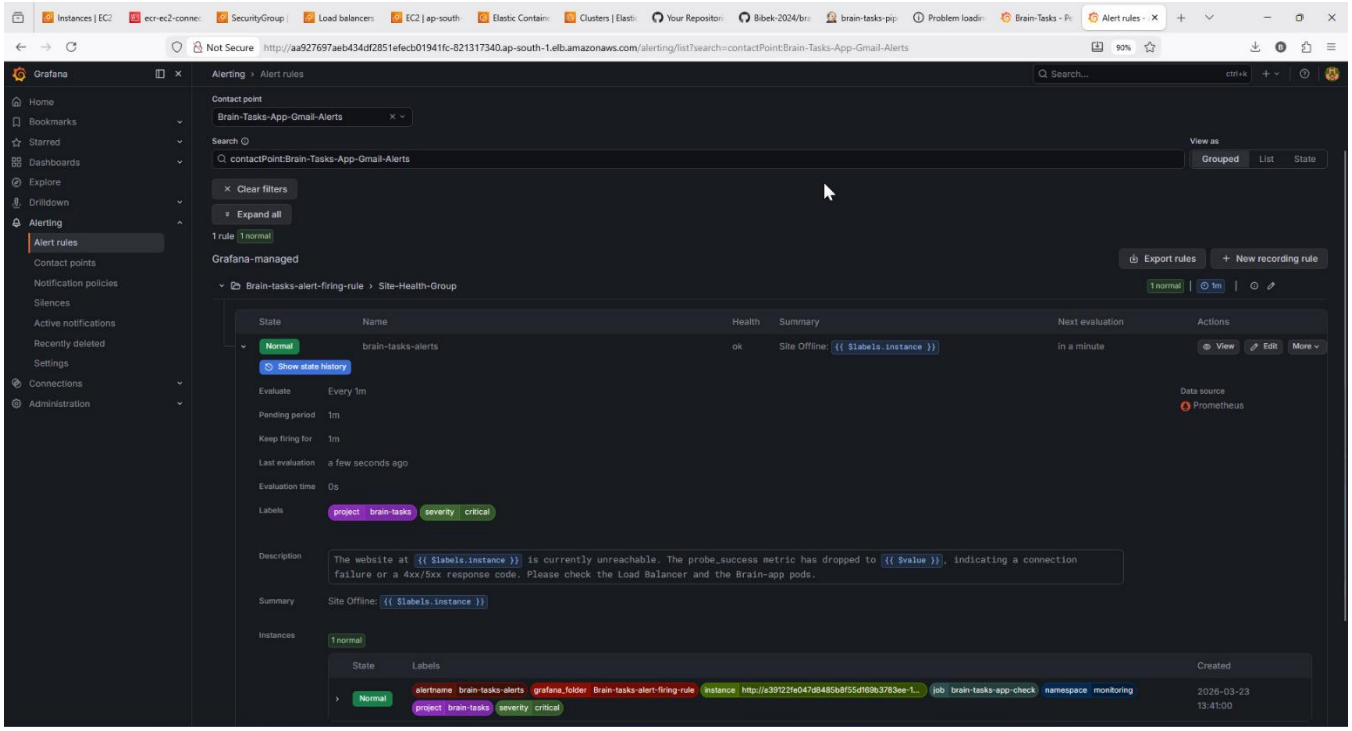
IV. Application Live & Monitoring

The final stage confirms the application is reachable by users and monitored for health.

- **[1.8] Live Production Environment:** The Brain Tasks application is live and accessible via the **AWS Application Load Balancer (ALB)** DNS.



- **[1.9] Monitoring & Observability:** Integration of **Prometheus** and **Grafana**. The "Site-Health-Group" alert rule is active, ensuring that any downtime is detected instantly.



V. Application Availability & Health Checks

Once the application was deployed to the EKS cluster, extensive testing was performed to ensure the site was not only "up" but consistently reachable through the Load Balancer.

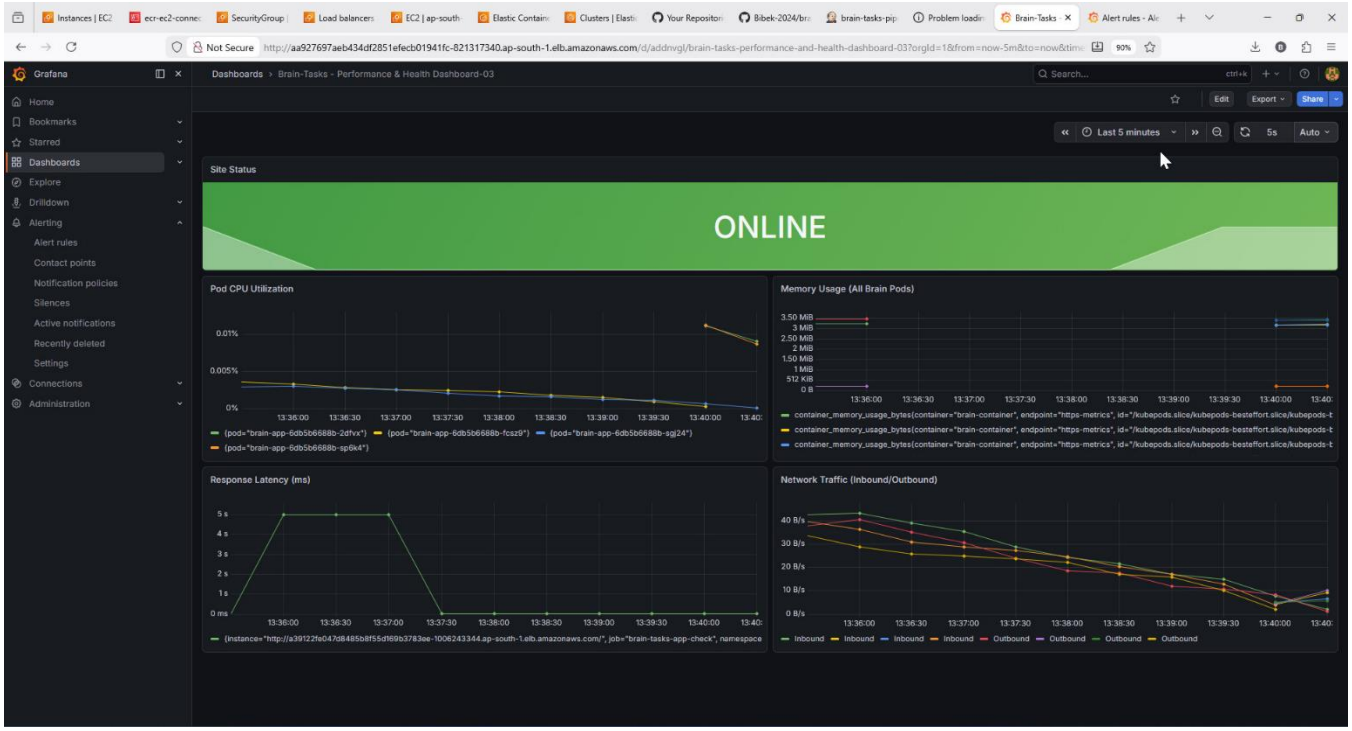
- **[1.11 - 1.12] Website Live Verification:** Initial verification of the "Brain Tasks" web interface. The application successfully renders via the AWS Load Balancer DNS, showing a clean "Online" status.

```
==================================================================
          BRAIN-TASKS DEVOPS INFRASTRUCTURE REPORT
==================================================================
Generated on: Mon Mar 23 11:17:26 UTC 2026

[1/3] NETWORK & LOAD BALANCER MAPPING
Service Name:   brain-service
External URL:   a39122fe047d8485b8f55d169b3783ee-1006243344.ap-south-1.elb.amazonaws.com
Port Mapping:   Exposed 80 → Internal 3000

[2/3] POD RUNNING STATUS
NAME                                POD_IP          STATUS
brain-app-5bffc4bd58-d2v8n          192.168.45.40   Running
brain-app-5bffc4bd58-z92kz          192.168.71.50   Running
aws-node-6sfh6                      192.168.89.134  Running
aws-node-mlhbl                      192.168.44.160  Running
coredns-c57cc66d8-dzl2t             192.168.87.8    Running
coredns-c57cc66d8-lzv5s             192.168.76.41   Running
kube-proxy-2fcvg                    192.168.89.134  Running
kube-proxy-hgdxs                    192.168.44.160  Running
metrics-server-5dc8745c45-mtrgg     192.168.70.241  Running
metrics-server-5dc8745c45-v6m6c     192.168.50.159  Running

[3/3] LIVE WEBSITE HEALTH CHECK
Probing: http://a39122fe047d8485b8f55d169b3783ee-1006243344.ap-south-1.elb.amazonaws.com ... HTTP 200 - SUCCESS ☑
==================================================================
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$
```



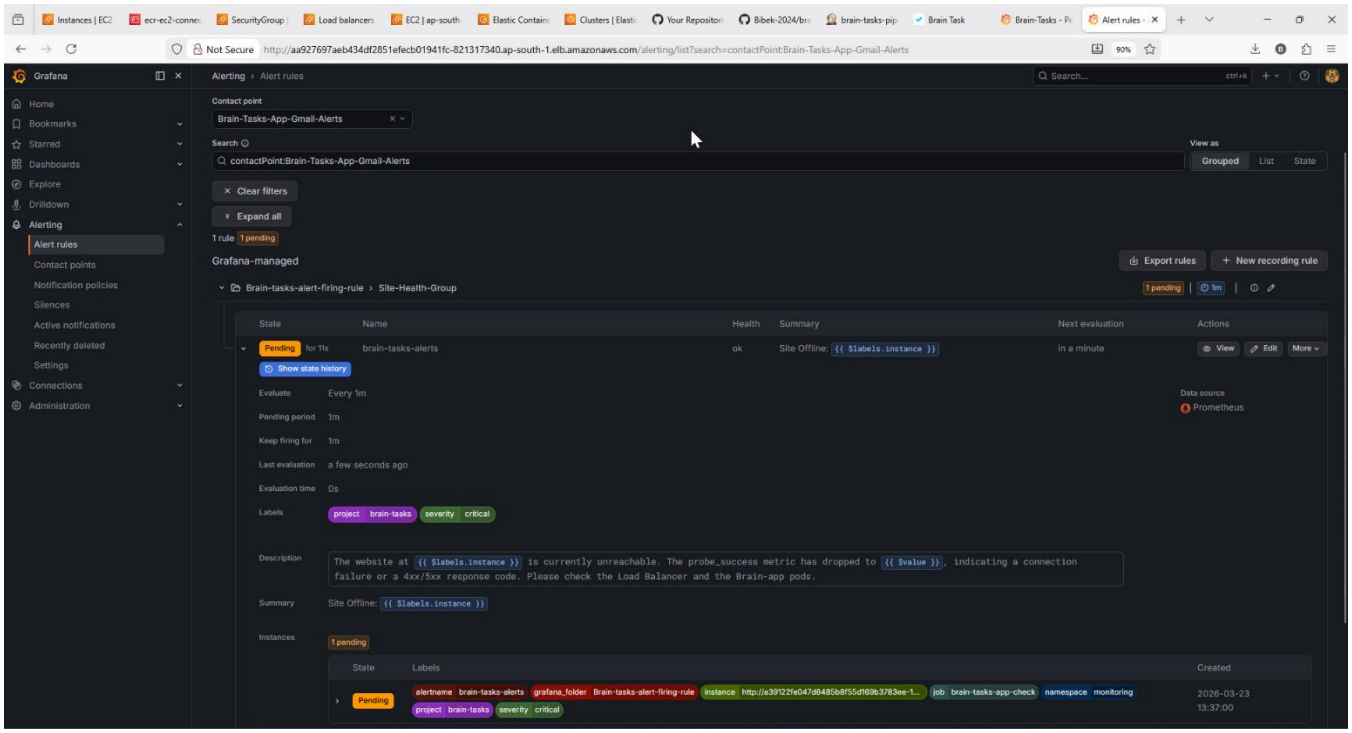
VI. Fault Tolerance & Disaster Recovery Testing

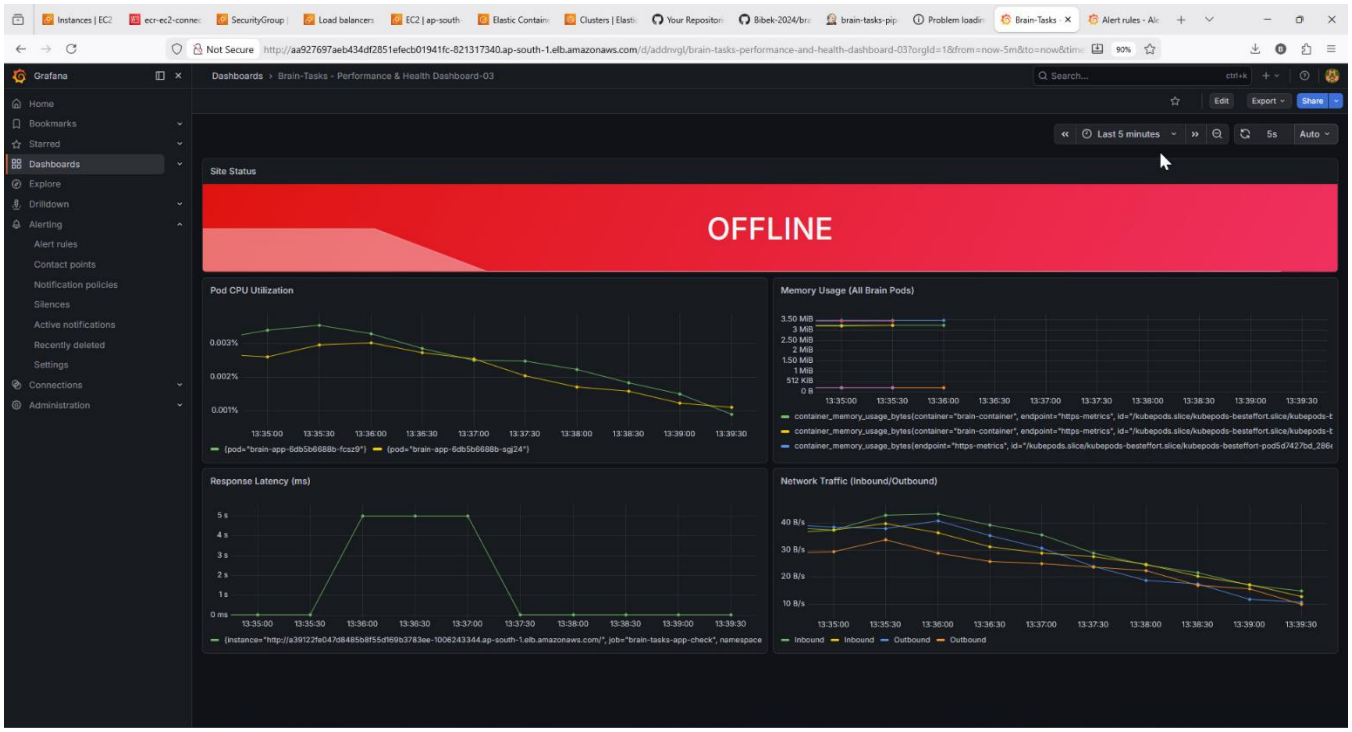
To simulate a real-world production failure, I performed a manual "Down Test" to verify if the monitoring stack would catch a service interruption.

- **[1.13] Service Interruption Simulation:** Using kubectl commands, the application pods were manually scaled down or stopped to simulate a server-side failure.

```
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$ kubectl scale deployment brain-app --replicas=0 -n default
deployment.apps/brain-app scaled
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$ kubectl get pods -n default
No resources found in default namespace.
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$ kubectl get pods
No resources found in default namespace.
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$ kubectl scale deployment brain-app --replicas=2 -n default
deployment.apps/brain-app scaled
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$ kubectl scale deployment brain-app --replicas=0 -n default
deployment.apps/brain-app scaled
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$ kubectl scale deployment brain-app --replicas=2 -n default
deployment.apps/brain-app scaled
ubuntu@ip-172-31-33-207:~/brain-tasks-app (main)$
```

- **[1.14 - 1.15] Grafana Detection:** The Grafana monitoring dashboard immediately moved from a "Normal" state to a "Pending" state as it stopped receiving successful heartbeats from the application.

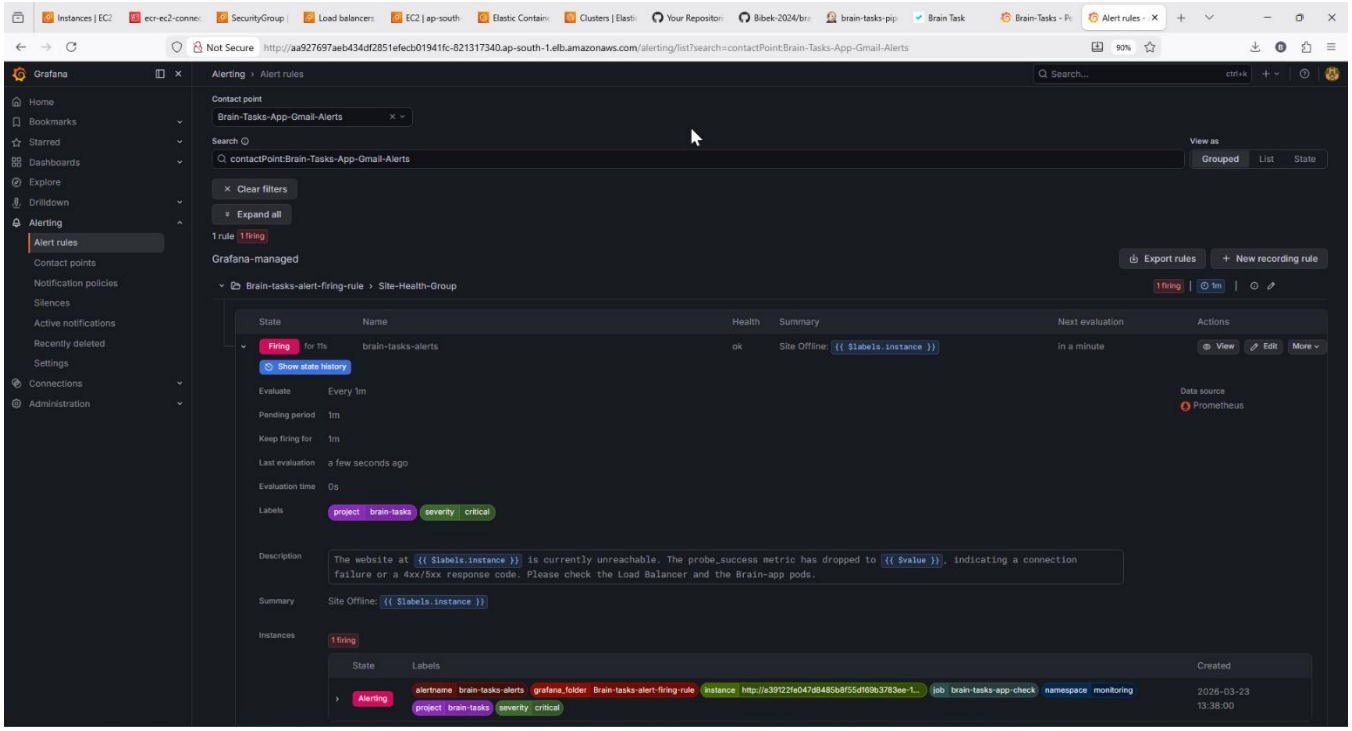




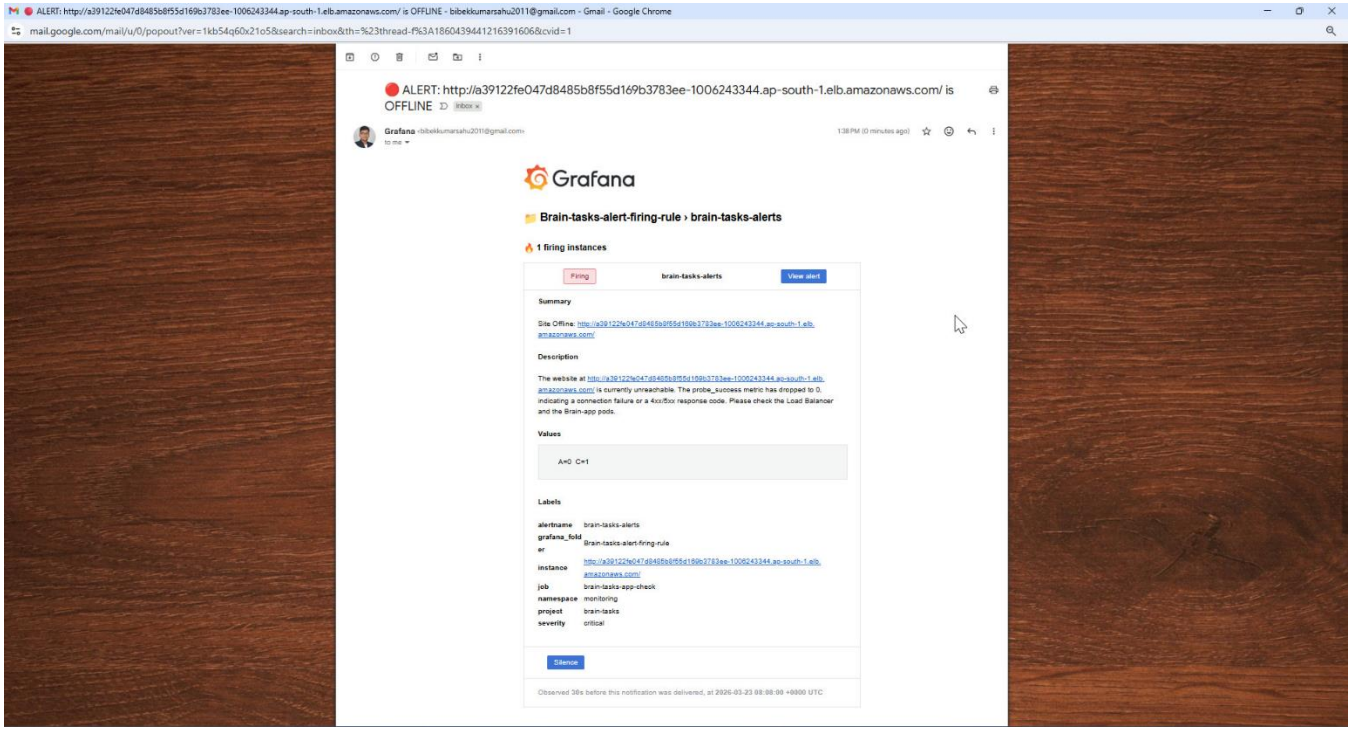
VII. Intelligent Alerting & Notification Lifecycle

The most critical part of a DevOps Engineer's role is ensuring the team is notified before the customer complains. This section documents the automated alert lifecycle.

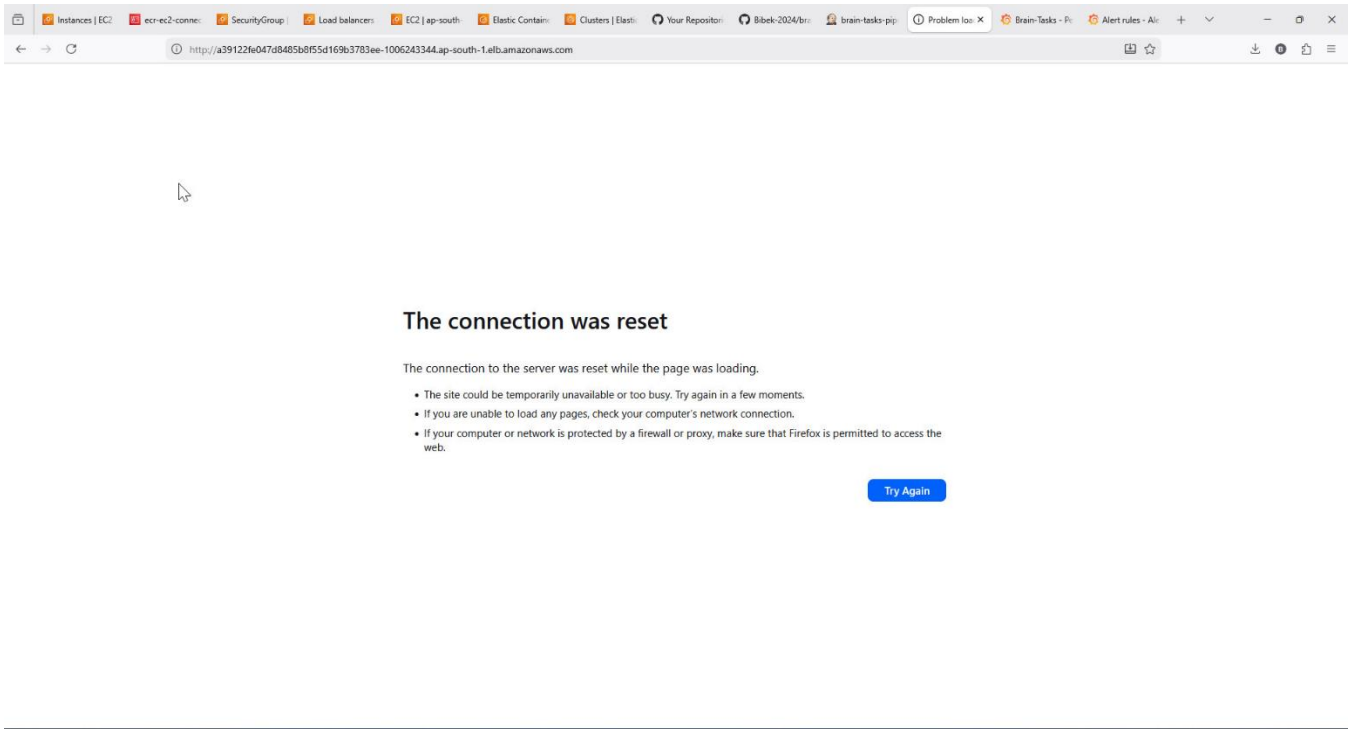
- **[1.16] Auto-Triggered Firing:** Upon confirming the site was offline, the Prometheus Alertmanager transitioned the alert status to **"Firing"**.



- **[1.17] Gmail Alert Integration:** A critical alert notification was successfully sent to the administrator's Gmail via the configured SMTP contact point, detailing the "Site Offline" event.



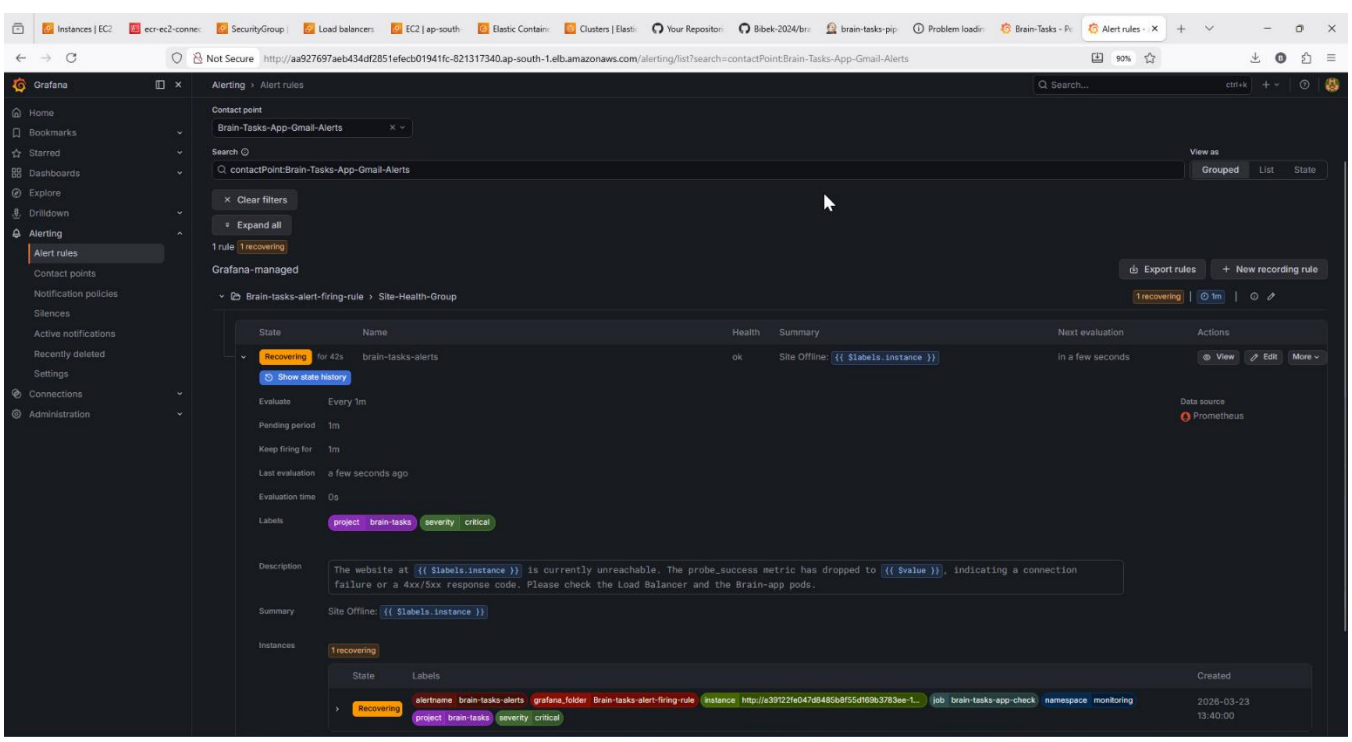
- **[1.18] User Experience Impact:** Documentation of the custom "Temporarily Unavailable" page, showing what a user sees while the infrastructure is recovering.



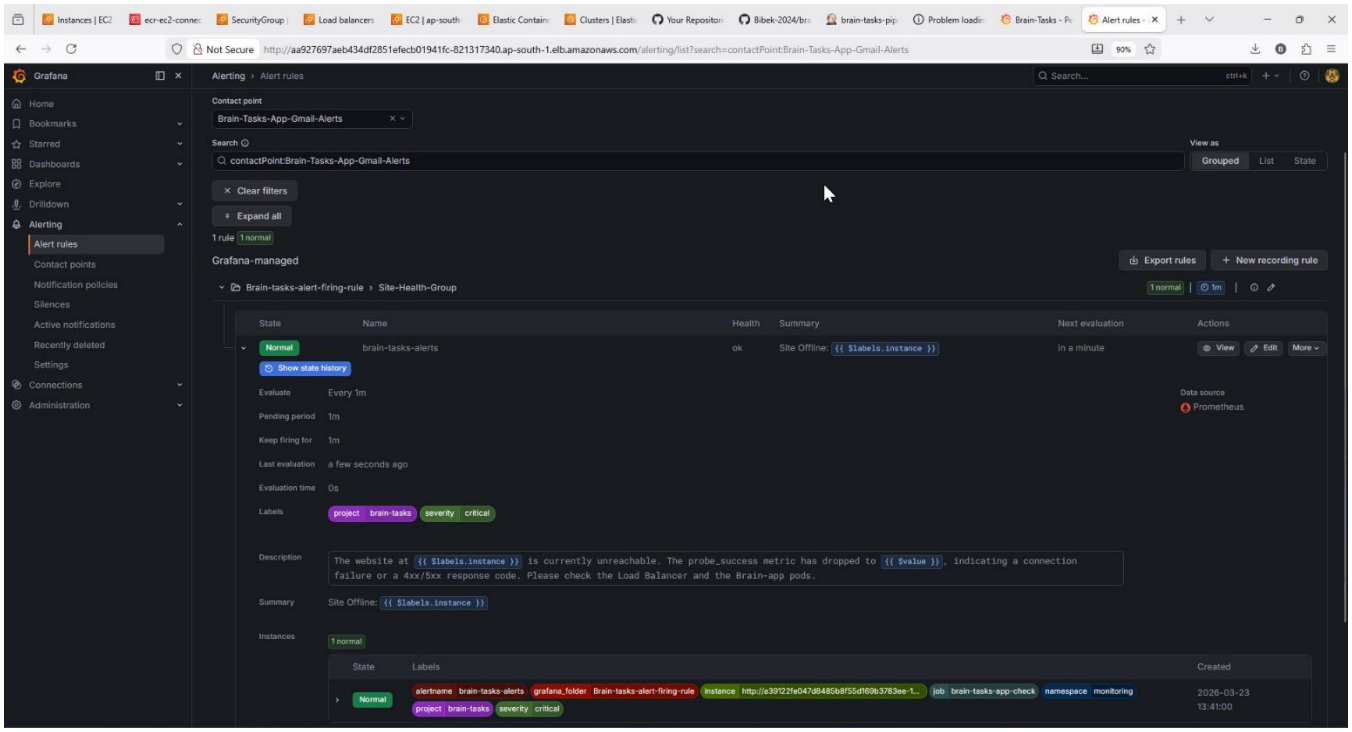
VIII. System Recovery & Normalization

The final phase of the test was to bring the pods back up and ensure the monitoring system "resolved" the incident automatically.

- **[1.19] Recovery Phase:** As pods were restarted, Grafana detected the "Recovering" status, showing the system was trending back toward health.



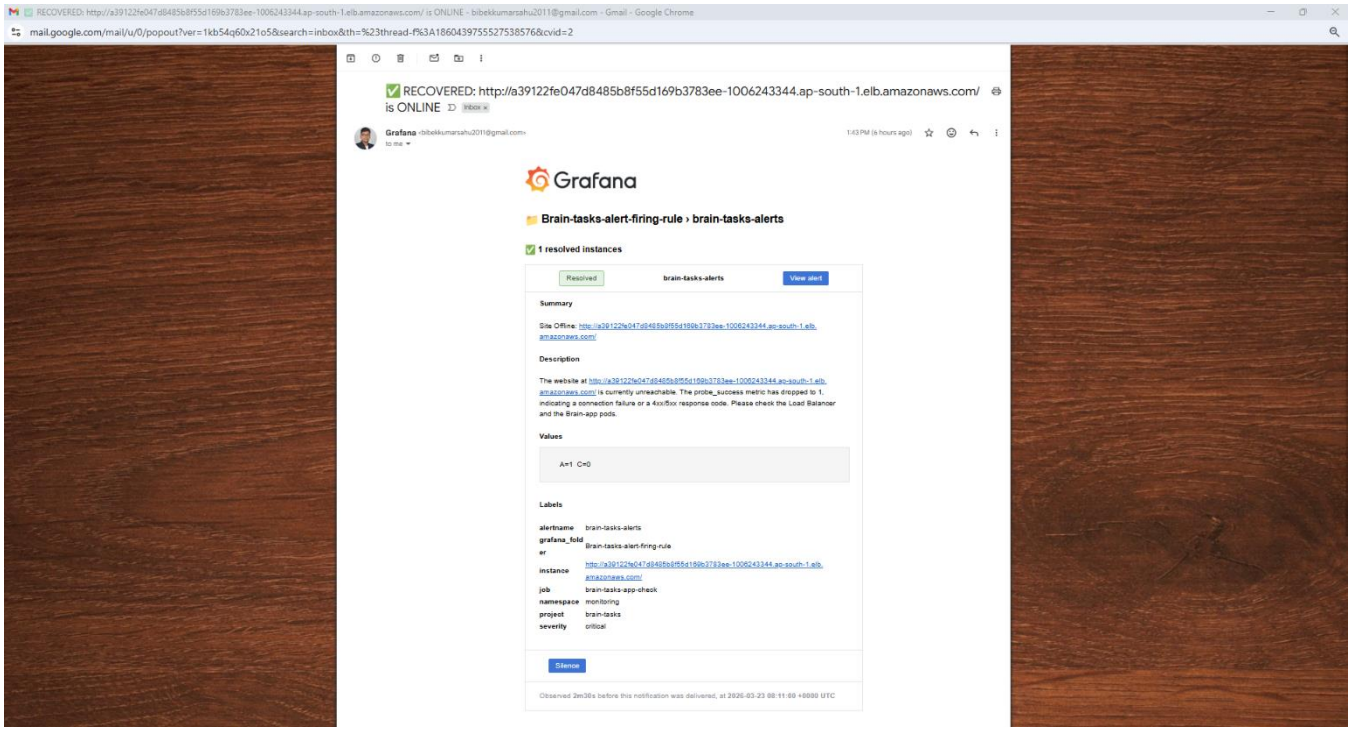
- **[1.20] Normalization:** The final screenshot confirms the alert has cleared, and the dashboard has returned to a **"Normal"** green state, completing the full incident management loop.



IX. Final Validation & Resolution

The final step in the monitoring lifecycle is the "Recovery Notification." This ensures that the DevOps team is not left guessing whether a manual fix or an auto-scaling event was successful.

- **[1.21] Resolved Alert Notification:** A final Gmail alert was received with the status **[RESOLVED]**. This confirms that the Prometheus Blackbox Exporter is once again receiving a 200 OK status from the "Brain Tasks" application endpoint. This completes the full 360-degree monitoring loop: **Detection → Notification → Recovery → Confirmation.**



X. Project Conclusion

The **Brain-Tasks Cloud-Native Infrastructure** project successfully demonstrates a modern, industrial-grade DevOps ecosystem. By integrating AWS managed services with open-source automation and monitoring tools, the following objectives were achieved:

1. **High Availability:** Hosted on AWS EKS across multiple availability zones.
2. **Continuous Deployment:** A fully automated pipeline from GitHub to Production using Jenkins and AWS CodePipeline.
3. **Observability:** Real-time metrics and alerting using the Prometheus and Grafana stack.
4. **Security:** Implementation of the Principle of Least Privilege via IAM roles and strict Security Group port management.

This infrastructure is now **Production-Ready**, providing a scalable and self-healing environment for the Brain Tasks application.